

ARTFM: Augmented Reality Visualization of Tool Functionality Manuals in Operating Rooms

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ABSTRACT

Error-free surgical procedures are crucial for a patient's health. However, with the increasing complexity and variety of surgical instruments, it is difficult for clinical staff to acquire detailed assembly and usage knowledge leading to errors in process and preparation steps. Yet, the gold standard in retrieving necessary information when problems occur is to get the paper-based manual. Reading through the necessary instructions is time-consuming and decreases care quality. We propose ARTFM, a process integrated manual, highlighting the correct parts needed, their location, and step-by-step instructions to combine the instrument using an augmented reality head-mounted display.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality

1 INTRODUCTION

Toady's processes in operating rooms rely on a multitude of different surgical tools. Keeping tabs on where everything is can be challenging, especially with similar looking instruments in turbulent situations. Some require complex, vendor specific handling. This makes it extremely challenging, even for experienced personnel, to ensure correct use and assembly. Therefore, paper manuals are often consulted, leading to increased assembly time and operation duration. A study showed that over 40% of surgical instrument defects are caused by inappropriate handling and over 15% by inadequate inspection [7]. Interviewing experts revealed that especially for individualized implants the assembly processes are often complex and highly specific, and thus a broad variety of surgical tools would need to be detected and supported by a digital assembly manual.

ARTFM tackles this crucial challenge by using an external camera and deep learning to locate and classify instruments and tools, providing the user with spatial highlights, contextual information, and assembly instructions using an augmented reality (AR) head-mounted display (HMD). Synthetic dataset generation aids the integration of manifold instruments. Our prototype conceptually approximates surgical tools with LEGO brick assemblies. The system can also act as an orientation guide for inexperienced nurses in a training procedure.

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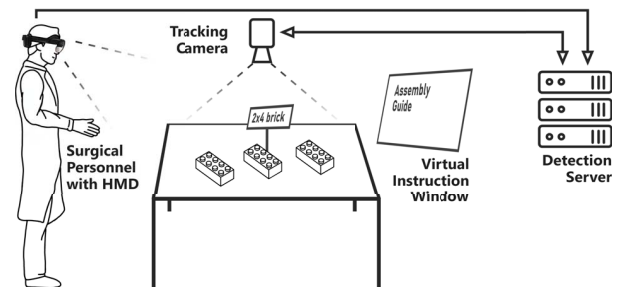


Figure 1: System principle with an instruction and highlighting.

2 RELATED WORK

Surgical instrument detection has been of wide interest, specifically in the area of minimally invasive surgery. Despite the success in various computer vision tasks with object detection and semantic segmentation, datasets for training a neural network with manifold current instruments are missing. Synthetic datasets have proven to be successful in filling this gap [2, 3] and provide the option to automate the inclusion of further instruments. Different approaches use game-engine or 3D rendering toolkits to programmatically create ground truth data for multiple methods like semantic segmentation, object detection, and 6D pose estimation.

AR on the other hand has been widely used for medical training or overlaying information during surgery. Context-aware AR instructions have successfully been used by others before, with abstract building blocks [5] or engineering assembly tasks [6] and were shown to be well suited for instructional content [1]. Yet, they have not been applied frequently in a surgical context per se. While the field of view and ergonomics need improvement, it's noted that assembly tasks benefit from AR instructions. Hands-free operation and ease of use were highlighted.

3 METHODOLOGY

Since 3D models of multi-part surgical instruments are often not publicly available, we chose LEGO brick assembly tasks for our proof of concept. We used *BlenderProc* [3] and *Unity Perception* [2] to generate our dataset consisting of 44 classes of bricks. The bricks are placed on a ground plate or in front of a crowded background, see Fig. 2a. To extend the data distribution and increase the network performance, parameters like position, orientation, color, lighting are randomized (domain randomization) and photorealistic, ray-tracing based rendering is used. To train the model 5000 images were generated. A training and evaluation split of 80:20 is used. For the object detection, we use Yolo5s [4], trained on our generated dataset. A learning rate of 0.001 was chosen using the Adam optimizer with a weight decay set to 1e-4. The batch size is set to 16. As mean average precision we report 84% on our evaluation split. To constantly

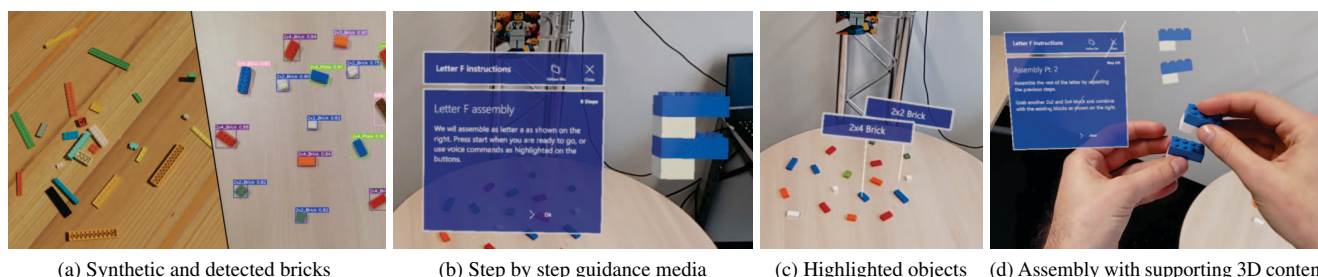


Figure 2: The user is supported with AR instructions, 3D media, and highlighted spatial information from detected objects.

monitor available LEGO bricks an Microsoft Azure Kinect camera is mounted approx. 50cm above a table. The object detection model can be called via RESTful API.

The application is built using Unity and is run on a HoloLens 2. To guide the user to objects' physical locations, the app fetches data from a server where the object detection model is executed. The alignment of the HMD and Kinect camera coordinate systems is achieved via image-based marker tracking (Vuuforia). Thus, objects do not need to be in the users' field of view to be detected. Objects are then highlighted using tooltips or a directional arrow when not in the users' central view. Instructions are organized into a set of steps, which are displayed in a freely movable information window. Three-dimensional animations are displayed beside when appropriate. These instructional animations support two-handed interaction to freely chose perspective and size. Through this visualization, the user can achieve an idea of the spatial composition of the individual components. The system supports the creation of new instruction with a baseline framework and state machine, easily extensible with custom animations and logic. Users can interact with the application via voice and gesture. The user's hands are monitored to detect input to buttons as well as interaction with physical objects.

4 PRELIMINARY EVALUATION

The prototype was evaluated internally by five colleagues aged between 25 and 30. Testers were not clinical staff, but had knowledge in the medical context. It was notable that users needed time to learn the new input modalities of HMDs. Our built in training mode can help users train assemblies while also familiarizing them with the device interaction. Some testers saw moved object locations due to short term tracking instabilities. The deep learning based object detection proved very accurate for LEGO bricks. Users noted that some expected usability features like back buttons are currently missing. The 3D objects and animations were seen as very helpful. Sometimes, important physical objects were overlaid by automatically placed windows, which could however be moved to a more pleasant position by the user. Overall, the testers had no problems interacting with the app after an initial exploration phase.

5 DISCUSSION

Our approach uses a synthetic dataset and a state-of-the-art object detection model to track objects and enable spatially aligned assembly instructions. We plan to explore calibration based on 6D pose estimation of the camera instead of marker-based tracking to reduce the setup complexity with commonly available hardware. This would also give us more control after the tracking process, allowing for a more purpose-built, stable, and precise tracking than the current solution. Photorealistic approaches and domain randomization were beneficial and should be expanded to improve results. In future works, we want to evaluate our system with surgical personnel. To this end, we plan to replace the currently used bricks with real

surgical objects to increase realism. We are positive our object detection approach will be sufficient as LEGO bricks are rather small, featureless and similarly shaped. However, the metallic surface of many surgical instruments may prove challenging to current object detection approaches. Another future feature will be the assembly phase detection to determine the current step in a manual using our synthetic data generation pipeline. This would allow consulting instructions for already half assembled tools and automatic instruction detection while requiring less attention and action from the user.

6 CONCLUSION

We present ARTFM, a prototype for an AR instruction manual for tool assembly during surgical procedures. Utilizing an underlying synthetic dataset generation pipeline enables an automated adaption of the deep learning model. As such, adding further objects or tools is straightforward, given their 3D models are available. Our approach provides spatial information and step-by-step in-situ AR instructions depending on the context, and could thus replace paper manuals in the OR in order to increase patient care quality.

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